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CONTENT OBTAINING METHOD, READABLE STORAGE MEDIUM, PROGRAM PRODUCT, AND ELECTRONIC DEVICE

Abstract

This application relates to the field of terminal technologies, and discloses a content-obtaining method, a readable storage medium, a program product, and an electronic device. When the electronic device detects that there is a security control on a current display interface, an area in which the security control is located is redrawn in a redrawing manner corresponding to each security control, to modify, clear, or replace user sensitive data, and retain display content corresponding to another control, so that a screenshot image or a video that does not include the user sensitive data is obtained.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/128962, filed on Nov. 1, 2023, which claims priority to Chinese Patent Application No. 202211493945.2, filed on Nov. 25, 2022. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD
[0002] This application relates to the field of terminal technologies, and in particular, to a content-obtaining method, a readable storage medium, a program product, and an electronic device.
BACKGROUND
[0003] To help a user save, as an image (through screenshot taking) or a video (through screen recording), an interface displayed by an electronic device, the electronic device may obtain, by using an application installed in the electronic device or an application or a service provided by an operating system of the electronic device, the interface displayed by the electronic device, and generate the corresponding image or the corresponding video. However, some display interfaces of the electronic device include data related to user privacy and security, for example, bank account information, a payment code, and identity information. If a screenshot is directly taken on the display interface, user data is easily leaked, and security of the user data is affected.
SUMMARY
[0004] In view of this, this application provides a content-obtaining method, a readable storage medium, a program product, and an electronic device. According to the method, a security control on a display interface of an electronic device is redrawn to obtain a screenshot image or a video. This not only meets a content-obtaining requirement (for example, screenshot taking or screen recording) of a user, but also protects user sensitive data on the display interface, thereby helping improve user experience.
[0005] According to a first aspect, an embodiment of this application provides a content-obtaining method, applied to an electronic device. The method includes: [0006] detecting a content-obtaining instruction of a user for a display interface on a screen of the electronic device; and [0007] in response to the content-obtaining instruction, generating display content data corresponding to the display interface, wherein when the display interface includes a first security control, the display content data includes first redrawn data obtained by redrawing the first security control in a first redrawing manner.
[0008] In other words, when the electronic device detects that the display interface to be obtained by the content-obtaining instruction includes a security control, the security control may be redrawn in a redrawing manner corresponding to the security control, to obtain the display content data corresponding to the content-obtaining instruction. In this way, even if there is a security control on a current display interface of the electronic device, a screenshot image or a video that does not include user sensitive data may be obtained after an area in which the security control is located is redrawn. This not only meets a content-obtaining requirement (for example, screenshot taking or screen recording) of the user, but also protects the user sensitive data on the display interface, thereby helping improve user experience.
[0009] In an implementation of the first aspect, when a security control identifier corresponding to

the first security control is valid, the electronic device determines that the display interface includes the first security control.
[0010] In some implementations, the security control identifier may be set in a layout file of an application corresponding to the first security control.
[0011] In an implementation of the first aspect, the security control identifier of the first security control is set to be valid when the layout file of the application corresponding to the first security control is loaded.
[0012] In other words, when the electronic device loads the application layout, the security control identifier on the display interface of the electronic device may be set to be valid, so that the electronic device may determine, based on the security control identifier, whether there is a security control on the display interface of the electronic device when performing screenshot taking or screen recording on the display interface of the electronic device.
[0013] In an implementation of the first aspect, the first redrawing manner is obtained from a layout file of a first application corresponding to the first security control.
[0014] Developers of the application may add a same screenshot redrawing strategy identifier or different screenshot redrawing strategy identifiers to the layout file for security controls, to control a manner of redrawing a security control by the electronic device when the display interface includes the security control, so that screenshot content obtained after the electronic device redraws the security control is similar to that on the display interface. This improves user experience.
[0015] In an implementation of the first aspect, the first redrawing manner includes at least one of the following: content clearing, filling, content replacing, hiding, and re-layout.
[0016] In an implementation of the first aspect, when the first redrawing manner is the content replacing, the redrawing the first security control in a first redrawing manner includes: [0017] obtaining a data type of user sensitive data in the first security control; and [0018] replacing the user sensitive data with preset data whose data type is the same as that of the user sensitive data.
[0019] The first security control is redrawn in a content replacing manner, so that a content attribute of the display interface can be retained as much as possible (improving a similarity between a screenshot image and the display interface), and the user sensitive data is not leaked. This helps further improve screenshot taking experience of the user.
[0020] In an implementation of the first aspect, when the first redrawing manner is the re-layout, the redrawing the first security control in a first redrawing manner includes: [0021] deleting a placeholder of the first security control in a current layout of an application corresponding to the first security control; and [0022] re-laying out another control in the current layout.
[0023] In other words, the first security control is redrawn in a re-layout manner, so that inharmonious layouts of controls in a screenshot image due to the placeholder of the security control can be avoided. This helps improve user experience.
[0024] In an implementation of the first aspect, when the first redrawing manner is the content clearing, the redrawing the first security control in a first redrawing manner includes: [0025] clearing user sensitive data in the first security control, and retaining a placeholder of the first security control in a current layout of an application corresponding to the first security control.
[0026] In an implementation of the first aspect, when the first redrawing manner is the filling, the redrawing the first security control in a first redrawing manner includes: [0027] filling an area in which user sensitive data in the first security control is located with any one or more of a preset color, a pattern, or a mosaic.
[0028] In an implementation of the first aspect, when the first redrawing manner is the hiding, the redrawing the first security control in a first redrawing manner includes: [0029] hiding the first security control, and retaining a placeholder of the first security control in a current layout of an application corresponding to the first security control.
[0030] In an implementation of the first aspect, the display interface further includes a second security control, and the display content data includes second redrawn data obtained by redrawing

the second security control in a second redrawing manner.
[0031] In other words, when there are a plurality of security controls on the display interface, the electronic device may redraw security controls in different redrawing manners for different security controls. This helps improve screenshot taking experience of the user.
[0032] In an implementation of the first aspect, the content-obtaining instruction includes a screenshot taking instruction or a screen recording instruction, and the display content data includes a screenshot image or a video.
[0033] In an implementation of the first aspect, an operating system of the electronic device is an Android™ system, and the Android system includes a graphics drawing module; and when it is determined that the display interface includes the first security control, the graphics drawing module generates the display content data corresponding to the content-obtaining instruction.
[0034] According to a second aspect, an embodiment of this application provides a computer-readable storage medium. The readable storage medium includes instructions. When the instructions are executed by an electronic device, the electronic device is enabled to implement the content-obtaining method provided in any one of the first aspect and the implementations of the first aspect.
[0035] According to a third aspect, an embodiment of this application provides an electronic device. The electronic device includes: a memory, where the memory stores instructions; and at least one processor, configured to execute the instructions, so that the electronic device implements the content-obtaining method provided in any one of the first aspect and the implementations of the first aspect.
[0036] According to a fourth aspect, an embodiment of this application provides a computer program product. When the computer program product runs on an electronic device, the electronic device is enabled to implement the content-obtaining method provided in any one of the first aspect and the implementations of the first aspect.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0037] FIG. **1**A is a diagram of a screenshot taking scenario according to some embodiments of this application;
[0038] FIG. **1**B is a diagram of another screenshot taking scenario according to some embodiments of this application;
[0039] FIG. **2** is a diagram of still another screenshot taking scenario according to some embodiments of this application;
[0040] FIG. **3**A is a diagram of a screenshot image obtained in a manner of filling an area of a security control with a grid pattern according to some embodiments of this application;
[0041] FIG. **3**B is a diagram of a screenshot image obtained in a manner of filling an area of a security control with a color according to some embodiments of this application;
[0042] FIG. **3**C is a diagram of a screenshot image obtained in a manner of filling an area of a security control with a mosaic according to some embodiments of this application;
[0043] FIG. **3**D is a diagram of a screenshot image obtained in a manner of replacing user sensitive data with data of a same type according to some embodiments of this application;
[0044] FIG. **3**E is a diagram of a screenshot image obtained in a manner of hiding a security control but retaining a placeholder according to some embodiments of this application;
[0045] FIG. **3**F is a diagram of a screenshot image obtained in a re-layout manner according to some embodiments of this application;
[0046] FIG. **3**G is a diagram of a screenshot image obtained by using different screenshot redrawing strategies for different security controls according to some embodiments of this

application;
[0047] FIG. **4**A is an flowchart of a content-obtaining method according to some embodiments of this application;
[0048] FIG. **4**B is a diagram of an interface displayed by a mobile phone after a screenshot of the interface including a security control is successfully taken according to some embodiments of this application;
[0049] FIG. **5** is a diagram of a software architecture of a mobile phone according to some embodiments of this application;
[0050] FIG. **6** is an flowchart of a content-obtaining method according to some embodiments of this application;
[0051] FIG. **7** is an interaction flowchart of a layout loading method according to some embodiments of this application;
[0052] FIG. **8**A and FIG. **8**B are an interaction flowchart of a content-obtaining method according to some embodiments of this application;
[0053] FIG. **9**A and FIG. **9**B are an interaction flowchart of another content-obtaining method according to some embodiments of this application; and
[0054] FIG. **10** is a diagram of a structure of a mobile phone according to some embodiments of this application.
DESCRIPTION OF EMBODIMENTS
[0055] The illustrative embodiments of this application include but are not limited to a content-obtaining method, a readable storage medium, a program product, and an electronic device.
[0056] It may be understood that technical solutions provided in embodiments of this application are applicable to any terminal device that can take a screenshot of a display interface of the terminal device, including but not limited to a mobile phone, a tablet computer, a wearable device (for example, a smartwatch or a smart band), an in-vehicle device (for example, an in-vehicle infotainment system), an internet of things device, a smart home device (for example, a smart television), and the like. For ease of description, the following describes the technical solutions of this application by using an example in which the terminal device is a mobile phone.
[0057] As described above, some display interfaces of the electronic device include data related to user privacy and security (referred to as user sensitive data below), for example, bank account information, a payment code, and user identity information. If a screenshot is directly taken on the display interface, user data is easily leaked, and security of the user data is affected.
[0058] Based on this, in some embodiments, a security window identifier (FLAG_SECURE) may be set for a window (referred to as a security window below) that is in an application in the electronic device and that includes the user sensitive data. In this way, when detecting a screenshot taking operation performed by a user, the electronic device may detect whether there is a window with a security window identifier on a current display interface of the electronic device, to determine whether the current display interface includes the user sensitive data. If the window includes a security window identifier, it indicates that the current display interface may include the user sensitive data, and screenshot taking or screen recording is not performed on the current display interface.
[0059] For example, FIG. **1**A is a diagram of a screenshot taking scenario according to some embodiments of this application. As shown in FIG. **1**A, a payment window **1** of a wallet application displayed by a mobile phone **10** includes a barcode control **11**, a two-dimensional code control **12**, and a payment account control **13**. The barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** record a barcode and a two-dimensional code that can be used to directly perform a transaction by using a bank card of the user, and bank card information of the user, and the wallet application may add a security window identifier for the payment window **1**. Therefore, when detecting a screenshot taking operation performed by the user, for example, when detecting an operation that the user knocks a screen of the mobile phone **10** with a knuckle, the

mobile phone **10** detects that there is a security window on a current display interface of the mobile phone **10**, namely, the payment window **1** of the wallet application, and further displays prompt information **14**, to prompt the user that “The current interface involves privacy content, and a screenshot is not allowed to be taken”, and not to take a screenshot of the current display interface of the mobile phone **10**. This avoids leakage of a payment code and bank card information of the user.

[0060] For another example, FIG. **1**B is a diagram of another screenshot taking scenario according to some embodiments of this application. As shown in FIG. **1**B, a mobile phone **10** performs screen recording on an interface displayed by the mobile phone **10**. At an 11.sup.th second during screen recording, the mobile phone **10** displays a payment window **1** of a wallet application. The payment window **1** of the wallet application is a security window. Therefore, the mobile phone **10** does not perform screen recording on a current display interface of the mobile phone **10**, but sets a video frame (a video frame corresponding to a video at the 11.sup.th second) corresponding to a screen recording result to all black. This avoids leakage of a payment code and bank card information of a user.

[0061] According to the foregoing method, leakage of user sensitive data can be effectively avoided. However, in some scenarios on an interface displayed by the user, only some controls in some windows are related to user sensitive data, and a purpose of taking a screenshot by the user is to store content other than the user sensitive data on the display interface of the electronic device. According to the foregoing method, once there is a security window on the current display interface of the electronic device, the electronic device does not take a screenshot of the current display interface, and therefore cannot obtain a screenshot image or a video required by the user, affecting the user experience.

[0062] In view of this, an embodiment of this application provides a content-obtaining method. An application or an operating system in an electronic device may set different screenshot redrawing strategies for a security control. In this way, when the electronic device detects that there is a security control on a current display interface, an area in which the security control is located is redrawn based on a screenshot redrawing strategy corresponding to each security control, and display content corresponding to another control is retained so that a screenshot image or a video that does not include user sensitive data is obtained. In this way, even if there is a security control on the current display interface of the electronic device, the screenshot image or the video that does not include the user sensitive data may be obtained after the area in which the security control is located is redrawn. This helps improve user experience.

[0063] For example, with reference to FIG. **2**, an interface of a wallet application displayed by a mobile phone **10** includes three security controls: a barcode control **11**, a two-dimensional code control **12**, and a payment account control **13**. When detecting a screenshot taking operation performed by a user, for example, when detecting an operation that the user knocks a screen of the mobile phone **10** with a knuckle, the mobile phone **10** detects that there is a security control on a current display interface of the mobile phone **10**. Then, the mobile phone **10** clears, based on the screenshot redrawing strategy corresponding to each security control, an area **11**S in which a payment barcode in the barcode control **11** is located, an area **12**S corresponding to a payment two-dimensional code in the two-dimensional code control **12**, and an area **13**S corresponding to a payment account in the payment account control **13**, to obtain a screenshot image. The screenshot image does not include the payment barcode, the payment two-dimensional code, or the payment account. This not only protects privacy data of the user, but also takes a screenshot of an area that is not related to the user sensitive data and that is on the interface of the wallet application, thereby improving the user experience.

[0064] It may be understood that, in some embodiments, the screenshot redrawing strategy may include at least one of the following processing strategies.

[0065] Content clearing: is to retain a placeholder of a security control on a display interface and clear content that is filled by an application and that is in the security control. For example, with reference to FIG. **2**, the mobile phone **10** may clear the payment barcode, the payment two-dimensional code, and the payment account that are filled by the application and that are respectively in the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13**.

[0066] Filling: is to retain a placeholder of a security control on a display interface and fill an area of the placeholder of the security control, for example, fill the area with any one or more of a pattern, a color, or a mosaic. For example, with reference to a screenshot image shown in FIG. **3**A, the mobile phone **10** may fill areas in which the original barcode control **11**, the original two-dimensional code control **12**, and the original payment account control **13** are located with a grid pattern. With reference to a screenshot image shown in FIG. **3**B, the mobile phone **10** may fill areas in which the original barcode control **11**, the original two-dimensional code control **12**, and the original payment account control **13** are located with a solid color. With reference to a screenshot image shown in FIG. **3**C, the mobile phone **10** may fill areas in which the original barcode control **11**, the original two-dimensional code control **12**, and the original payment account control **13** are located with a mosaic.

[0067] Content replacing: is to retain a placeholder of a security control on a display interface and replace user sensitive data with another data (for example, another data whose data type is the same as or different from that in the security control) in the security control. For example, with reference to a screenshot image shown in FIG. **3**D, the mobile phone **10** may replace the payment barcode in the barcode control **11** with a barcode whose indication content is "123456789", replace the payment two-dimensional code in the two-dimensional code control **12** with a two-dimensional code whose indication content is "123456789", and replace the payment account information in the payment account control **13** with "******". Based on the screenshot redrawing strategy, a content attribute of the current display interface of the mobile phone **10** can be retained as much as possible (improving a similarity between a screenshot image and the display interface), and the user sensitive data is not leaked. This helps further improve the screenshot taking experience of the user. It may be understood that, in some embodiments, after the user sensitive data is replaced with another data whose data type is the same as that in the security control, an identifier indicating that content in an area of the security control has been replaced may be further added to remind the user.

[0068] Hiding: is to retain a placeholder of a security control on a display interface but hide the interface of the security control. For example, with reference to FIG. **3**E, the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** are all displayed in the payment window **1**, and a card swiping control **22** is displayed at a lower layer of the payment window **1**. The mobile phone **10** may hide the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** so that the payment window **1** no longer includes any control, the payment window **1** is hidden, all content of the card swiping control **22** is displayed in the screenshot image, and locations of other controls on the interface keep unchanged.

[0069] Re-layout: is to delete a placeholder of a security control on a display interface and re-lay out an interface of an application corresponding to the security control. For example, with reference to a screenshot image shown in FIG. **3**F, the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** are all displayed in the payment window **1** and a card swiping control **22** is displayed at a lower layer of the payment window **1**. The mobile phone **10** may delete placeholders of the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** so that the payment window **1** no longer includes any control, a placeholder of the payment window **1** is deleted, all content of the card swiping control **22** is displayed in the screenshot image, and a location of a control **23** is moved upward. This avoids inharmonious layouts of controls in the screenshot image due to placeholders of each security control and helps improve user experience.

[0070] It may be understood that, in some embodiments, for some controls having a high security requirement, or in a case in which a security control is displayed in a full-screen manner, the screenshot redrawing strategy may further include screenshot prohibiting, namely, not allowing the mobile phone **10** to take a screenshot of the current display interface.
[0071] It may be understood that, in some other embodiments, another screenshot redrawing strategy may be further included. This is not limited herein.
[0072] It may be understood that, in some embodiments, different security controls of a same application may have different screenshot redrawing strategies. For example, with reference to a screenshot image shown in FIG. **3**G, the mobile phone **10** may clear a payment barcode in an area **11**S in which the barcode control **11** is located, fill an area **12**S in which the two-dimensional code control **12** is located with a grid pattern, and fill an area **13**S in which the payment account control **13** is located with a mosaic.
[0073] It may be understood that, in some other embodiments, the screenshot redrawing strategy may further include any strategy that may be used to modify, delete, or replace the user sensitive data. This is not limited herein.
[0074] The following further describes the technical solutions of this application with reference to the cases shown in FIG. **2** to FIG. **3**G.
[0075] FIG. **4**A is an flowchart of a content-obtaining method according to some embodiments of this application. The method is performed by the mobile phone **10**. As shown in FIG. **4**, the procedure includes the following steps: [0076] S**401**: Detect a screenshot taking instruction.
[0077] The mobile phone **10** detects the screenshot taking instruction to trigger the content-obtaining method provided in embodiments of this application.
[0078] For example, when the mobile phone **10** detects a screenshot taking gesture, a screenshot taking operation, or a screenshot taking voice defined by an operating system, a screenshot taking application or service, a screen recording application or service, or the like of the mobile phone **10**, the screenshot taking instruction may be detected, to trigger the content-obtaining method provided in embodiments of this application.
[0079] For example, the screenshot taking gesture may include but is not limited to an operation that the user knocks a display of the mobile phone **10** with a knuckle, a multi-finger swiping operation performed by the user on the display of the mobile phone **10**, and the like.
[0080] For another example, the screenshot taking operation may include but is not limited to a triggering operation performed by the user on a screenshot taking control in a menu bar of the mobile phone **10** or an application of the mobile phone **10**, a preset operation on a volume control or a power control of the mobile phone **10**, and the like.
[0081] For another example, the screenshot taking voice may include a preset screenshot taking voice detected by the operating system or an installed application in the mobile phone **10**. For example, when capturing a voice including “screenshot taking”, “image capturing”, or “screen recording”, the mobile phone **10** may detect the screenshot taking instruction.
[0082] It may be understood that, in some other embodiments, the mobile phone **10** may also detect the screenshot taking instruction in another case. A source and a form of the screenshot taking instruction are not limited herein. [0083] S**402**: Determine whether there is a security control on a current display interface.
[0084] After detecting the screenshot taking instruction, the mobile phone **10** determines whether there is a security control on the current display interface of the mobile phone **10**. If there is a security control on the current display interface, it indicates that the current display interface may include user sensitive data, and step S**403** is performed. Otherwise, it indicates that the current display interface does not include user sensitive data, and step S**405** is performed.
[0085] For example, in some embodiments, an application or the operating system in the mobile phone **10** may attach a security control mark to a control. When a security control mark of a control is valid, it indicates that the control is a security control. Otherwise, it indicates that the control is

not a security control. In this way, the mobile phone **10** may: obtain a security control mark of each control on the current display interface, and when a security control mark of a control is valid, determine that there is a security control on the current display interface; or when the security control mark of each control on the current display interface is invalid, determine that there is no security control on the current display interface. A specific manner of setting the security control mark is described below with reference to a software architecture of the mobile phone **10**. Details are not described herein again.
[0086] It may be understood that, in some other embodiments, the mobile phone **10** may alternatively determine, in another manner, whether there is a security control on the current display interface. This is not limited herein. For example, in some embodiments, the mobile phone **10** may record an identifier of the security control. [0087] S**403**: Obtain a screenshot redrawing strategy of each security control on the current display interface.
[0088] When determining that there is a security control on the current display interface, the mobile phone **10** obtains the screenshot redrawing strategy corresponding to each security control.
[0089] For example, in some embodiments, each application in the mobile phone **10** presets a corresponding screenshot redrawing strategy for a security control included in each application, and the mobile phone **10** may obtain, from an application corresponding to each security control, the screenshot redrawing strategy corresponding to each security control.
[0090] It may be understood that, in some embodiments, the mobile phone **10** may further set corresponding screenshot redrawing strategies for different types of security controls, and obtain, based on a type of each security control, a screenshot redrawing strategy corresponding to each security control, so that the mobile phone **10** may query, from the mobile phone **10** based on the type of each security control, the screenshot redrawing strategy corresponding to each security control.
[0091] It may be understood that, the screenshot redrawing strategy may include but is not limited to content clearing, filling, content replacing, hiding, re-layout, screenshot prohibiting, and the like that are described above. [0092] S**404**: Redraw, based on the screenshot redrawing strategy of each security control, an area in which each security control on the current display interface is located, to obtain a screenshot image.
[0093] After obtaining the screenshot redrawing strategy of each security control, the mobile phone **10** redraws, based on the screenshot redrawing strategy corresponding to each security control, the area corresponding to each security control on the current display interface of the mobile phone **10**, to obtain the screenshot image.
[0094] For example, in the case shown in FIG. **2**, the mobile phone **10** may set the areas corresponding to the security controls to be transparent, to obtain the screenshot image. For another example, in the cases shown in FIG. **3**A to FIG. **3**C, the mobile phone **10** may fill the areas corresponding to the security controls with any one or more of the pattern, the color, or the mosaic, to obtain the screenshot image. For another example, in the case shown in FIG. **3**D, the mobile phone **10** may replace the payment barcode in the barcode control **11** with the barcode whose indication content is “123456789”, replace the payment two-dimensional code in the two-dimensional code control **12** with the two-dimensional code whose indication content is “123456789”, and replace the payment account information in the payment account control **13** with “******”, to obtain the screenshot image. For another example, in the case shown in FIG. **3**E, the mobile phone **10** may retain the placeholder of the security control and hide the security control, to obtain the screenshot image. For another example, in the case shown in FIG. **3**F, the mobile phone **10** may hide the security control and re-lay out the display interface of the application (the wallet application) on the current display interface, to obtain the screenshot image. For another example, in the case shown in FIG. **3**G, the mobile phone **10** may clear the barcode in the barcode control, fill the area **12**S corresponding to the two-dimensional code control **12** with the grid pattern, and fill the area **13**S corresponding to the payment account control **13** with the mosaic to obtain the

screenshot image.

[0095] It may be understood that, in some embodiments, when the generated screenshot image is an image obtained after the security control is redrawn, the mobile phone **10** may further prompt the user that the user sensitive data has been processed in the screenshot image. For example, with reference to FIG. **4**B, in the case shown in FIG. F, after generating the screenshot image, the mobile phone **10** may further display a screenshot image **42** on the current display interface of the mobile phone **10** in an overlay manner, and display a prompt information box **41** to prompt the user that "A screenshot is successfully taken, and user sensitive data has been anonymized". [0096] S**405**: Directly generate a screenshot image corresponding to the current display interface.

[0097] When there is no security control on the current display interface, the mobile phone **10** directly draws the current display interface as an image, to obtain the screenshot image corresponding to the current display interface.

[0098] According to the content-obtaining method provided in embodiments of this application, the electronic device may take a screenshot of the display interface including the security control so that a screenshot taking requirement of the user can be met. In addition, because an area that is in the screenshot image and that includes the user sensitive data is redrawn based on a corresponding screenshot redrawing strategy, leakage of the user sensitive data is avoided. In addition, the electronic device may further use different screenshot redrawing strategies for different security controls. This improves user experience.

[0099] The following further describes the technical solutions in embodiments of this application with reference to a software architecture of the mobile phone **10**.

[0100] For ease of understanding, the software architecture of the mobile phone **10** is first described.

[0101] It may be understood that the software architecture of the mobile phone **10** may be any software architecture such as iOS™, Android™, or HarmonyOS™. The following uses Android as an example to describe the software architecture of the mobile phone **10**.

[0102] FIG. **5** is a diagram of a structure of the software architecture according to some embodiments of this application.

[0103] As shown in FIG. **5**, the software architecture of the mobile phone **10** is a multi-layer structure, including an application layer **01**, an application framework layer **02** (referred to as a framework layer for short), a native layer (native) **03**, and a driver layer **04**.

[0104] The application layer includes applications in the mobile phone **10**, for example, a desktop, a user interface, setting, a screenshot taking application **011**, a screen recording application **012**, and another application **013**.

[0105] The screenshot taking application **011** may be configured to: in response to a screenshot taking operation performed by a user, take a screenshot of a current display interface of the mobile phone **10** by using interfaces and services of the framework layer **02**, the native layer **03**, and the driver layer **04**, to obtain a screenshot image; and save the screenshot image or share the screenshot image to another electronic device by using another application. The screen recording application **012** may be configured to: in response to a screenshot taking operation performed by the user, take a screenshot of a current display interface of the mobile phone **10** by using interfaces and services of the framework layer **02**, the native layer **03**, and the driver layer **04**, to obtain a screenshot video; and save the screenshot video or share the screenshot video to another electronic device by using another application.

[0106] It may be understood that, in some other embodiments, the application layer **01** may include more or fewer applications. This is not limited herein.

[0107] The application framework layer **02** is configured to provide an application programming interface (API) and a programming framework for an application at the application layer **01**. For example, the application framework layer **02** may include a window manager (WindowManagerService, WMS) **021**, a gesture recognizer (MotionRecognition) **022**, a control

023 (UiKit), a view manager (View) **024**, a graphics control (SurfaceControl) **025**, a knuckle recognizer (HwGestureAction) **026**, a security control (SecUikit) **027**, and the like.

[0108] An application manager is a system service provided by an Android™ system, and is configured to manage startup and loading of an application and a service in the mobile phone **10**.

[0109] The window manager **021** is configured to manage a window in the mobile phone **10**, for example, monitor a life cycle of the window, an input and a focus event, a screen direction, conversion, animation, a location, and deformation. The window manager **021** is further configured to send all window metadata to a graphics drawer (SurfaceFlinger) **032**, so that the graphics drawer **032** can synthesize the data on a screen into a user interface.

[0110] The gesture recognizer **022** is configured to recognize a gesture of the user acting on a touchscreen of the mobile phone **10**, for example, sliding, tapping, and pinching.

[0111] The control **023** is a unit used for layout in an application, and a user interface displayed in the application is obtained by combining and overlaying a plurality of controls. In some embodiments, an attribute, a combination relationship, an overlay relationship, a location relationship, and the like of each control on each user interface of the application may be recorded in a layout file.

[0112] The view manager **024** is configured to obtain, based on the layout file of the application, the attribute, the combination relationship, the location relationship, the overlay relationship, and the like of each control on the user interface of the application. In some embodiments, the view manager **024** may be further configured to: record a screenshot redrawing strategy identifier and a screenshot redrawing strategy that correspond to a security control on the user interface of the application, and adjust each security attribute of the application based on the screenshot redrawing strategy, to redraw the security control.

[0113] The graphics control **025** is configured to control and manage a corresponding control in a window of each application in the mobile phone **10**, for example, creation, destruction, transparency, display, or hiding of each control.

[0114] The knuckle recognizer **026** is configured to recognize whether the user acts on the touchscreen of the mobile phone **10** with a knuckle.

[0115] The security control **027** is a control whose security control mark is valid in an attribute of the control. An application may include a control and a security control. In some embodiments, attribute information of the security control may further include a screenshot redrawing strategy identifier corresponding to the security control, and the screenshot redrawing strategy identifier indicates a screenshot redrawing strategy for redrawing the security control when the electronic device takes a screenshot of the security control.

[0116] In some embodiments, the security control mark and the screenshot redrawing strategy identifier of the security control may be embedded in the layout file of the application, or may be set by setting a corresponding interface by using the security control. A setting form, a storage form, and the like of the security control mark and the screenshot redrawing strategy identifier are not limited herein.

[0117] It may be understood that, in some other embodiments, the application framework layer **02** may include more or fewer modules or services. This is not limited herein.

[0118] The native layer **03** includes some common native services, link libraries, and the like. The native layer may communicate with upper-layer Java code (referred to as a JNI mechanism) in C and C++ languages, and may further interact with a bottom-layer hardware driver. In some embodiments, the native layer may include a knuckle gesture (KnuckleGesture) **031**, the graphics drawer (SurfaceFlinger) **032**, and the like.

[0119] The knuckle gesture **031** is configured to recognize a gesture of the user acting on the touchscreen of the mobile phone **10** with a knuckle, for example, a single-knuckle knocking gesture or a dual-knuckle knocking gesture. In this way, the screenshot taking application **011** or the screen recording application **012** may detect a screenshot taking instruction when the knuckle gesture **031**

recognizes the single-knuckle knocking gesture or the dual-knuckle knocking gesture.
[0120] The graphics drawer **032** is configured to draw application windows, controls, and the like in the mobile phone **10** as images for display on a display, or generate a screenshot image.
[0121] In particular, in some embodiments, the graphics drawer **032** is further configured to: when receiving a request for drawing the screenshot image, determine whether there is a security control on a current display interface of the mobile phone **10**; and when determining that there is a security control, send a notification message indicating that there is a security control on the current display interface to the graphics control **025**, to notify a control identifier and the like of the security control on the current display interface.
[0122] It may be understood that the control identifier is an identifier, for example, an identity (ID) or a control name, that may uniquely indicate a control.
[0123] It may be understood that, in some embodiments, the graphics drawer **032** may alternatively be disposed at the application framework layer **02**. This is not limited herein.
[0124] It may be understood that, in some other embodiments, the native layer **03** may include more or fewer modules or services. This is not limited herein.
[0125] The driver layer **04** may include drivers of hardware and modules of the mobile phone **10**, for example, a touchscreen driver **041** and a display driver **042**.
[0126] Based on the software architecture shown in FIG. **5**, with reference to FIG. **6**, after detecting a screenshot taking operation triggered by the user, the mobile phone **10** may send a screenshot taking instruction to the screenshot taking application **011** or the screen recording application **012** (the screenshot taking application **011** is used as an example below). The screenshot taking application **011** sends a screenshot taking request to the graphics control **025**. After receiving the screenshot taking request, the graphics control **025** forwards the screenshot taking request to the graphics drawer **032**. After receiving the screenshot taking request, the graphics drawer **032** determines whether there is a security control on the current display interface of the mobile phone **10**.
[0127] If there is a security control on the current display interface of the mobile phone **10**, the graphics drawer **032** notifies an application corresponding to the security control to perform redrawing (re-draw an attribute of a control in a window of a new application and set a layout). The application corresponding to the security control sends a redrawing request to the view manager **024** to trigger redrawing. In response to the redrawing request, the view manager **024** adjusts an attribute and a layout of a control on the current display interface based on a screenshot redrawing strategy corresponding to the security control, to delete, modify, or replace user sensitive data on the current display interface. Then, the graphics drawer **032** generates a screenshot image based on a re-layout and redrawn interface and sends the screenshot image to the screenshot taking application **011**. Finally, the screenshot taking application **11** stores or shares the screenshot image.
[0128] If there is no security control on the current display interface of the mobile phone **10**, the graphics drawer **032** directly generates a screenshot image based on the current display interface of the mobile phone **10** and sends the screenshot image to the screenshot taking application **011**, and then the screenshot taking application **011** stores or shares the screenshot image.
[0129] In this way, when there is a security control on the current display interface of the mobile phone **10**, that is, when the current display interface includes the user sensitive data, a screenshot may be taken on the current display interface of the mobile phone **10**. This helps improve user experience. In addition, the user sensitive data in the generated screenshot image is deleted, modified, or replaced based on the corresponding redrawing strategy so that leakage of the user sensitive data can be avoided and security of user data can be improved.
[0130] The following further describes the technical solutions of this application with reference to the software architecture shown in FIG. **5**.
[0131] As described above, in the software architecture shown in FIG. **5**, the graphics drawer **032** determines whether there is a security control on the current display interface of the mobile phone

10. Therefore, a security control mark of the security control needs to be set to be valid in the graphics drawer **032** so that the graphics drawer **032** may determine, based on whether a security identifier of each control is valid, whether each control is a security control.
[0132] FIG. **7** is an interaction flowchart of a layout loading method according to some embodiments of this application.
[0133] S**701**: Send a layout file to a view manager **024**.
[0134] During startup, an application sends the layout file to the view manager **024**.
[0135] S**702**: Parse the layout file and record a control identifier of a security control and a corresponding screenshot redrawing strategy identifier.
[0136] After receiving the layout file sent by the application, the view manager **024** parses the layout file to obtain a security control mark of each control in a window of the application and determines control identifiers of security controls whose security control mark is valid. Then, the control identifier of the security control and the corresponding screenshot redrawing strategy identifier are recorded.
[0137] It may be understood that a screenshot redrawing strategy identifier of a security control indicates a screenshot redrawing strategy of the security control during screenshot taking.
[0138] For example, in some embodiments, a screenshot redrawing strategy identifier corresponding to the screenshot redrawing strategy “content clearing” may be “empty”; a screenshot regrouping strategy identifier corresponding to the screenshot redrawing strategy “filling” may be “mask”; a screenshot regrouping strategy identifier corresponding to the screenshot redrawing strategy “content replacing” may be “replace”; a screenshot regrouping strategy identifier corresponding to the screenshot redrawing strategy “hiding” may be “invisible”; a screenshot regrouping strategy identifier corresponding to the screenshot redrawing strategy “re-layout” may be “re-layout”; and a screenshot regrouping strategy identifier corresponding to the screenshot redrawing strategy “screenshot prohibiting” may be “forbidden”.
[0139] It may be understood that the screenshot redrawing strategy identifier and the screenshot redrawing strategy are merely examples. In some other embodiments, another screenshot redrawing strategy identifier and another screenshot redrawing strategy may be used. This is not limited herein.
[0140] S**703**: The view manager **024** sends the control identifier of the security control to a graphics control **025**.
[0141] The view manager **024** sends the control identifier of the security control to the graphics control **025**.
[0142] S**704**: The graphics control **025** sets a security control mark of the security control to be valid, and sends the control identifier of the security control to a graphics drawer **032**.
[0143] After receiving the control identifier of the security control, the graphics control **025** adds a valid security control identifier to the control corresponding to the control identifier and sends the control identifier of the security control to the graphics drawer **032**.
[0144] S**705**: Draw a display interface and set the security control mark of the security control to be valid.
[0145] The graphics drawer **032** draws the display interface, and adds the valid security control identifier for the control corresponding to the control identifier. For example, in the case shown in FIG. **2**, the graphics drawer **032** of the mobile phone **10** may draw a payment interface of the wallet application and add valid security control identifiers for the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13**.
[0146] According to the method provided in embodiments of this application, when the electronic device loads the application layout, the graphics drawer **032** may set the security control identifier on the display interface of the electronic device to be valid so that the electronic device may determine, based on the security control identifier, whether there is a security control on the display interface of the electronic device when performing screenshot taking or screen recording on the

display interface of the electronic device.

[0147] In addition, developers of the application may add a same screenshot redrawing strategy identifier or different screenshot redrawing strategy identifiers to the layout file for security controls, to control a manner of redrawing a security control by the electronic device when the display interface includes the security control, so that screenshot content obtained after the electronic device redraws the security control is similar to that on the display interface. This improves user experience.

[0148] In addition, because the mobile phone **10** generates a screenshot image or a video by using the graphics drawer **032**, when the application layout is loaded, the graphics drawer **032** adds the valid security control identifier for the security control on the drawn display interface so that the graphics drawer **032** can directly determine whether there is a security control on the display interface of the electronic device, and does not interact with the video manager **024** or the application to determine whether there is a security control on the display interface. This helps improve a speed of generating the screenshot image or the video by the mobile phone **10**.

[0149] It may be understood that after adding the valid security control identifier for the security control on the current display interface of the mobile phone **10**, the graphics drawer **032** may take, according to the content-obtaining method provided in embodiments of this application, a screenshot of the display interface including the security control.

[0150] FIG. **8**A and FIG. **8**B are an interaction flowchart of a content-obtaining method according to some embodiments of this application.

[0151] As shown in FIG. **8**A and FIG. **8**B, the interaction procedure includes the following steps:

[0152] S**801**: A screenshot taking application **011** detects a screenshot taking instruction and sends a screenshot taking request to a graphics control **025**.

[0153] After detecting the screenshot taking instruction, the screenshot taking application **011** sends the screenshot taking request to the graphics control **025**.

[0154] For example, when a mobile phone **10** detects a screenshot taking gesture, a screenshot taking operation, or a screenshot taking voice defined by an operating system, a screenshot taking application or service, a screen recording application or service, or the like of the mobile phone **10**, the screenshot taking application **011** may detect the screenshot taking instruction and send the screenshot taking request to the graphics control **025**.

[0155] For example, in the case shown in FIG. **2**, when recognizing a single-knuckle knocking operation performed by a user on a touchscreen of the mobile phone **10**, a knuckle gesture **031** may send a screenshot taking instruction to the screenshot taking application **011**. After detecting the screenshot taking instruction, the screenshot taking application **011** sends a screenshot taking request to the graphics control **025**.

[0156] It may be understood that, for content of the screenshot taking gesture, the screenshot taking operation, and the screenshot taking voice, refer to the step S**401**. Details are not described herein again. [0157] S**802**: The graphics control **025** sends a screenshot image drawing request to a graphics drawer **032**.

[0158] After receiving the screenshot taking request sent by the screenshot taking application **011**, the graphics control **025** sends the screenshot image drawing request to the graphics drawer **032** in response to the screenshot taking request. [0159] S**803**: The graphics drawer **032** determines whether there is a security control on a current interface.

[0160] The graphics drawer **032** determines, in response to the screenshot image drawing request, whether there is a security control on the current display interface of the mobile phone **10**. If there is a security control on the current display interface, it indicates that the current display interface may include user sensitive data, and step S**805** is performed. Otherwise, it indicates that the current display interface does not include user sensitive data, and step S**804** is performed.

[0161] For example, the graphics drawer **032** may: obtain a security control mark of each control on the current display interface of the mobile phone **10**, and when a security control mark of a control is valid, determine that there is a security control on the current display interface, and go to step S**805**; or when the security control mark of each control on the current display interface is invalid, determine that there is no security control on the current display interface, and go to step S**804**. [0162] S**804**: The graphics drawer **032** draws a screenshot image based on the current display interface and sends the screenshot image to the screenshot taking application **011**.

[0163] When there is no security control on the current display interface of the mobile phone **10**, the graphics drawer **032** directly draws a screenshot image based on the current display interface and sends the drawn screenshot image to the screenshot taking application **011**.

[0164] It may be understood that, in some embodiments, the graphics drawer **032** may first send the screenshot image to the graphics control **025**, and then the graphics control **025** sends the screenshot image to the screenshot taking application **011**. [0165] S**805**: The graphics drawer **032** sends a control identifier of the security control.

[0166] When determining that the current display interface includes the security control, the graphics drawer **032** sends the control identifier of the security control to the view manager **024**.

[0167] For example, in the case shown in FIG. **2**, the graphics drawer **032** may send control identifiers **11**, **12**, and **13** of the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** to the view manager **024**. [0168] S**806**: Determine, based on the security control identifier, a screenshot redrawing strategy corresponding to the security control, and send a re-layout and redrawing request to a corresponding application.

[0169] After receiving the control identifier of the security control, the view manager **024** queries a screenshot redrawing strategy identifier corresponding to each security control, and sends the re-layout and redrawing request to the corresponding application based on a screenshot redrawing strategy corresponding to the screenshot redrawing strategy identifier.

[0170] For example, when determining that a screenshot redrawing strategy identifier corresponding to a security control is “empty”, the view manager **024** sends, to an application corresponding to the security control, a re-layout and redrawing request for clearing content that is filled by the application and that is in the security control. When determining that a screenshot redrawing strategy identifier corresponding to a security control is “mask”, the view manager **024** sends, to an application corresponding to the security control, a re-layout and redrawing request for filling an area corresponding to the security control with any one or more of a pattern, a color, a mosaic, or the like. When determining that a screenshot redrawing strategy identifier corresponding to a security control is “replace”, the view manager **024** sends, to an application corresponding to the security control, a re-layout and redrawing request for replacing user sensitive data with data whose data type is the same as that of the user sensitive data in the security control. When determining that a screenshot redrawing strategy identifier corresponding to a security control is “invisible”, the view manager **024** sends, to an application corresponding to the security control, a re-layout and redrawing request for retaining a placeholder of the security control but hiding the security control. When determining that a screenshot redrawing strategy identifier corresponding to a security control is “re-layout”, the view manager **024** sends, to an application corresponding to the security control, a re-layout and redrawing request for deleting a placeholder of the security control and re-laying out a window of the application. [0171] S**807**: The application corresponding to the security control sends the re-layout and redrawing request to the view manager **024**.

[0172] After receiving the re-layout and redrawing request, the application corresponding to the security control forwards the re-layout and redrawing request to the view manager **024**. [0173] S**808**: The view manager **024** re-lays out the current interface and sends the re-drawing request to the graphics drawer **032**.

[0174] In response to the re-layout and redrawing request, the view manager **024** adjusts a layout attribute of each control based on a screenshot redrawing strategy corresponding to each security control, for example, whether to hide, whether to delete, and a location, to re-lay out each control. In addition, the view manager **024** sends the redrawing request to the graphics drawer **032** after the

re-layout.
[0175] For example, in the case shown in FIG. **2**, the view manager **024** may clear the payment barcode, the payment two-dimensional code, and the payment account that are filled by the application and that are respectively in the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13**. In the cases shown in FIG. **3**A to FIG. **3**C, the view manager **024** may fill the areas corresponding to the payment barcode, the payment two-dimensional code, and the payment account in the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** with a pattern (for example, a grid pattern), a color, and a mosaic respectively. In the case shown in FIG. **3**D, the view manager **024** may replace the payment barcode in the barcode control **11** with the barcode whose indication content is "123456789", replace the payment two-dimensional code in the two-dimensional code control **12** with a two-dimensional code whose indication content is "123456789", and replace the payment account information in the payment account control **13** with "******". In the case shown in FIG. **3**E, the view manager **024** may set attributes of the barcode control **11**, the two-dimensional code control **12**, the payment account control **13**, and the payment window **1** to be hidden. In the case shown in FIG. **3**F, the view manager **024** may delete placeholders of the barcode control **11**, the two-dimensional code control **12**, the payment account control **13**, and the payment window **1**, and adjust a location of the control **23**. [0176] S**809**: The graphics drawer **032** draws a screenshot image based on an adjusted layout and sends the screenshot image to the graphics control **025**.
[0177] The graphics drawer **032** draws the screenshot image based on the adjusted layout and sends the screenshot image to the graphics control **025**. [0178] S**810**: The graphics control **025** sends the screenshot image to the screenshot taking application **011**.
[0179] After receiving the screenshot image, the graphics control **025** sends the screenshot image to the screenshot taking application **011**.
[0180] It may be understood that after receiving the screenshot image, the screenshot taking application **011** may store the screenshot image as a picture or share the screenshot image to another electronic device.
[0181] It may be understood that, when screen recording is performed on the mobile phone **10** by using the screen recording application **012**, each frame of image during screen recording may be obtained according to the steps S**801** to S**810**. Details are not described herein again.
[0182] According to the content-obtaining method provided in embodiments of this application, the mobile phone **10** may take a screenshot of the current display interface of the mobile phone **10** when the current display interface includes the security control. This improves user experience. In addition, because the security control is redrawn based on the corresponding screenshot redrawing strategy, the screenshot image obtained after screenshot taking does not include user privacy information. This helps improve data security of the user. Moreover, because different security controls may use different screenshot redrawing strategies, a variety of the screenshot image may be increased. This helps improve user experience.
[0183] It may be understood that the technical solutions of this application described based on the software architecture shown in FIG. **5** are merely examples. In some other embodiments, functions of some modules may be combined or split, or functions of the modules shown in FIG. **5** may be performed by another module, or functions of some modules may be replaced. This is not limited herein.
[0184] It may be understood that, in some embodiments, an execution process of the steps S**801** to S**810** is merely an example. In some other embodiments, an execution sequence and an execution body of some steps may be adjusted, or some steps may be combined or split. This is not limited herein.
[0185] For example, in some embodiments, after receiving the control identifier of the security control, the view manager **024** may directly adjust the layout based on the screenshot redrawing strategy corresponding to each security control and send the redrawing request to the graphics

drawer **032** instead of initiating the re-layout and redrawing request by the application corresponding to the security control.
[0186] FIG. **9**A and FIG. **9**B are an interaction flowchart of another content-obtaining method according to some embodiments of this application. As shown in FIG. **9**A and FIG. **9**B, the interaction procedure includes the following steps: [0187] S**901**: A screenshot taking application **011** detects a screenshot taking instruction and sends a screenshot taking request to a graphics control **025**.
[0188] After detecting the screenshot taking instruction, the screenshot taking application **011** sends the screenshot taking request to the graphics control **025**. For details, refer to the step S**801**. Details are not described herein again. [0189] S**902**: The graphics control **025** sends a screenshot image drawing request to a graphics drawer **032**.
[0190] After receiving the screenshot taking request sent by the screenshot taking application **011**, the graphics control **025** sends the screenshot image drawing request to the graphics drawer **032** in response to the screenshot taking request. [0191] S**903**: The graphics drawer **032** determines whether there is a security control on a current interface.
[0192] The graphics drawer **032** determines, in response to the screenshot image drawing request, whether there is a security control on the current display interface of the mobile phone **10**. If there is a security control on the current display interface, it indicates that the current display interface may include user sensitive data, and step S**905** is performed. Otherwise, it indicates that the current display interface does not include user sensitive data, and step S**904** is performed. For a manner of determining whether there is a security control on the current interface, refer to the step S**803**. Details are not described herein again. [0193] S**904**: The graphics drawer **032** draws a screenshot image based on the current display interface and sends the screenshot image to the screenshot taking application **011**.
[0194] When there is no security control on the current display interface of the mobile phone **10**, the graphics drawer **032** directly draws a screenshot image based on the current display interface and sends the drawn screenshot image to the screenshot taking application **011**.
[0195] It may be understood that, in some embodiments, the graphics drawer **032** may first send the screenshot image to the graphics control **025**, and then the graphics control **025** sends the screenshot image to the screenshot taking application **011**. [0196] S**905**: The graphics drawer **032** sends a control identifier of the security control.
[0197] When determining that the current display interface includes the security control, the graphics drawer **032** sends the control identifier of the security control to the view manager **024**.
[0198] For example, in the case shown in FIG. **2**, the graphics drawer **032** may send control identifiers **11**, **12**, and **13** of the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** to the view manager **024**. [0199] S**906**: Determine, based on the security control identifier, a screenshot redrawing strategy corresponding to each security control and adjust a layout based on the screenshot redrawing strategy corresponding to each security control.
[0200] After receiving the control identifier of the security control, the view manager **024** queries a screenshot redrawing strategy identifier corresponding to each security control and lays out the current display interface based on the screenshot redrawing strategy corresponding to the screenshot redrawing strategy identifier, for example, adjusts a layout attribute of the control, for example, whether to hide, whether to delete, a location, and content filling, to re-lay out each control.
[0201] For example, in the case shown in FIG. **2**, the view manager **024** may clear the payment barcode, the payment two-dimensional code, and the payment account that are filled by the application and that are respectively in the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13**. In the cases shown in FIG. **3**A to FIG. **3**C, the view manager **024** may fill the areas corresponding to the payment barcode, the payment two-

dimensional code, and the payment account in the barcode control **11**, the two-dimensional code control **12**, and the payment account control **13** with a pattern (for example, a grid pattern), a color, and a mosaic respectively. In the case shown in FIG. **3**D, the view manager **024** may replace the payment barcode in the barcode control **11** with the barcode whose indication content is "123456789", replace the payment two-dimensional code in the two-dimensional code control **12** with a two-dimensional code whose indication content is "123456789", and replace the payment account information in the payment account control **13** with "******". In the case shown in FIG. **3**E, the view manager **024** may set attributes of the barcode control **11**, the two-dimensional code control **12**, the payment account control **13**, and the payment window **1** to be hidden. In the case shown in FIG. **3**F, the view manager **024** may delete placeholders of the barcode control **11**, the two-dimensional code control **12**, the payment account control **13**, and the payment window **1**, and adjust a location of the control **23**. [0202] S**907**: The view manager **024** sends a redrawing request to the graphics drawer **032**.

[0203] The view manager **024** sends the redrawing request to the graphics drawer **032** after the re-layout. [0204] S**908**: The graphics drawer **032** draws a screenshot image based on an adjusted layout and sends the screenshot image to the graphics control **025**.

[0205] The graphics drawer **032** draws the screenshot image based on the adjusted layout and sends the screenshot image to the graphics control **025**. [0206] S**909**: The graphics control **025** sends the screenshot image to the screenshot taking application **011**.

[0207] After receiving the screenshot image, the graphics control **025** sends the screenshot image to the screenshot taking application **011**.

[0208] According to the content-obtaining method provided in embodiments of this application, the mobile phone **10** may take a screenshot of the current display interface of the mobile phone **10** when the current display interface includes the security control. This improves the user experience. In addition, because the security control is redrawn based on the corresponding screenshot redrawing strategy, the screenshot image obtained after screenshot taking does not include user privacy information. This helps improve data security of the user. Moreover, because different security controls may use different screenshot redrawing strategies, a variety of the screenshot image may be increased. This helps improve user experience.

[0209] Further, FIG. **10** is a diagram of a structure of a mobile phone **10** according to some embodiments of this application.

[0210] As shown in FIG. **10**, the mobile phone **10** may include a processor **110**, an external memory interface **120**, an internal memory **121**, a universal serial bus (USB) interface **130**, a charging management module **140**, a power management module **141**, a battery **142**, an antenna **1**, an antenna **2**, a mobile communication module **150**, a wireless communication module **160**, an audio module **170**, a speaker **170**A, a receiver **170**B, a microphone **170**C, a headset jack **170**D, a sensor module **180**, a button **190**, a motor **191**, an indicator **192**, a camera **193**, a display **194**, a subscriber identity module (SIM) card interface **195**, a communication apparatus **196**, and the like. The sensor module **180** may include a pressure sensor **180**A, a gyroscope sensor **180**B, a barometric pressure sensor **180**C, a magnetic sensor **180**D, an acceleration sensor **180**E, a distance sensor **180**F, an optical proximity sensor **180**G, a fingerprint sensor **180**H, a temperature sensor **180**J, a touch sensor **180**K, an ambient light sensor **180**L, a bone conduction sensor **180**M, and the like.

[0211] The processor **110** may include one or more processing units. For example, the processor **110** may include an application processor (AP), a modem processor, a graphics processing unit (GPU), an image signal processor (ISP), a controller, a video codec, a digital signal processor (DSP), a baseband processor, and/or a neural-network processing unit (NPU). Different processing units may be independent devices, or may be integrated into one or more processors.

[0212] The controller may generate an operation control signal based on an instruction operation code and a time sequence signal, to complete control of instruction reading and instruction execution.

[0213] A memory may be further disposed in the processor **110**, and is configured to store instructions and data. In some embodiments, the memory in the processor **110** is a cache memory. The memory may store instructions or data just used or cyclically used by the processor **110**. If the processor **110** needs to use the instructions or the data again, the processor **110** may directly invoke the instructions or the data from the memory. This avoids repeated access, reduces waiting time of the processor **110**, and improves system efficiency.

[0214] In some embodiments, the processor **110** may be configured to: execute instructions corresponding to the content-obtaining method provided in embodiments of this application; and when a screenshot taking operation performed by a user is detected, generate a screenshot image that does not include user sensitive data.

[0215] The USB interface **130** is an interface that conforms to a USB standard specification, and may be a mini USB interface, a micro USB interface, a USB type-C interface, or the like. The USB interface **130** may be configured to connect to a charger to charge the mobile phone **10**, or may be configured to exchange data between the mobile phone **10** and a peripheral device, or may be configured to connect to a headset for playing an audio through the headset. The interface may be further configured to connect to another electronic device such as an AR device.

[0216] The charging management module **140** is configured to receive a charging input from a charger. When charging the battery **142**, the charging management module **140** may further supply power to the electronic device by using the power management module **141**.

[0217] The power management module **141** is configured to connect to the battery **142**, the charging management module **140**, and the processor **110**. The power management module **141** receives an input from the battery **142** and/or the charging management module **140**, and supplies power to the processor **110**, the internal memory **121**, the display **194**, the camera **193**, the wireless communication module **160**, and the like.

[0218] A wireless communication function of the mobile phone **10** may be implemented using the antenna **1**, the antenna **2**, the mobile communication module **150**, the wireless communication module **160**, the modem processor, the baseband processor, and the like.

[0219] The antenna **1** and the antenna **2** are configured to transmit and receive an electromagnetic wave signal.

[0220] The mobile communication module **150** may provide a wireless communication solution that includes 2G/3G/4G/5G or the like and that is applied to the mobile phone **10**. The mobile communication module **150** may include at least one filter, a switch, a power amplifier, a low noise amplifier (LNA), and the like. The mobile communication module **150** may receive an electromagnetic wave through the antenna **1**, perform processing such as filtering or amplification on the received electromagnetic wave, and send a processed electromagnetic wave to the modem processor for demodulation. The mobile communication module **150** may further amplify a signal modulated by the modem processor and convert the signal into an electromagnetic wave for radiation through the antenna **1**. In some embodiments, at least some functional modules in the mobile communication module **150** may be disposed in the processor **110**. In some embodiments, at least some functional modules of the mobile communication module **150** may be disposed in a same device as at least some modules of the processor **110**.

[0221] The wireless communication module **160** may provide a wireless communication solution that is applied to the mobile phone **10** and that includes a wireless local area network (WLAN) (for example, a wireless fidelity (Wi-Fi) network), Bluetooth™ (BT), a global navigation satellite system (GNSS), frequency modulation (FM), near field communication (NFC), an infrared (IR) technology, and the like. The wireless communication module **160** may be one or more devices integrating at least one communication processing module. The wireless communication module **160** may receive an electromagnetic wave through the antenna **2**, perform processing such as filtering or amplification on the received electromagnetic wave, and send a processed electromagnetic wave to the modem processor for demodulation. The wireless communication module **160** may further amplify a signal modulated by the modem processor, and convert the signal into an electromagnetic wave for radiation through the antenna **2**.

[0222] The mobile phone **10** implements a display function by using the GPU, the display **194**, the application processor, and the like. The GPU is a microprocessor for image processing, and is connected to the display **194** and the application processor. The GPU is configured to: perform mathematical and geometric computation, and render an image. The processor **110** may include one or more GPUs, which execute program instructions to generate or change display information.

[0223] The display **194** is configured to display an image, a video, and the like. The display **194** includes a display panel. The display panel may be a liquid crystal display (LCD), an organic light-emitting diode (OLED), an active-matrix organic light-emitting diode (AMOLED), a flexible light-emitting diode (FLED), a mini-LED, a micro-LED, a micro-OLED, a quantum dot light-emitting diode (QLED), or the like. In some embodiments, the mobile phone **10** may include one or N displays **194**, where N is a positive integer greater than 1.

[0224] The camera **193** is configured to capture a static image or a video. An optical image of an object is generated through a lens, and is projected onto a photosensitive element. The photosensitive element may be a charge-coupled device (CCD) or a complementary metal-oxide-semiconductor (CMOS) phototransistor. The photosensitive element converts an optical signal into an electrical signal and then transmits the electrical signal to the ISP to convert the electrical signal into a digital image signal. The ISP outputs the digital image signal to the DSP for processing. The DSP converts the digital image signal into an image signal in a standard format such as RGB or YUV. In some embodiments, the mobile phone **10** may include one or N cameras **193**, where Nis a positive integer greater than 1.

[0225] The external memory interface **120** may be configured to connect to an external storage card, for example, a micro SD card, to extend a storage capability of the mobile phone **10**. The external memory card communicates with the processor **110** through the external memory interface **120** to implement a data storage function. For example, files such as music and videos are stored in the external storage card.

[0226] The internal memory **121** may be configured to store computer-executable program code, where the executable program code includes instructions, for example, the memory **103**. The internal memory **121** may include a program storage area and a data storage area. The program storage area may store an operating system, an application required by at least one function, and the like. The data storage area may store data and the like created in a process of using the mobile phone **10**, for example, may be used to store a control identifier of a security control, a screenshot redrawing strategy identifier corresponding to the security control, and a pattern, an image, a text, and the like corresponding to a screenshot redrawing strategy. In addition, the internal memory **121** may include a high-speed random access memory, or may include a non-volatile memory, for example, at least one magnetic disk storage device, a flash memory, or a universal flash storage (UFS). The processor **110** executes the instructions stored in the internal memory **121** and/or the instructions stored in the memory disposed in the processor **110** to perform various function applications of the mobile phone **10**.

[0227] The mobile phone **10** may implement an audio function, for example, music playing or recording, by using the audio module **170**, the speaker **170**A, the receiver **170**B, the microphone **170**C, the headset jack **170**D, the application processor, and the like.

[0228] The audio module **170** is configured to convert digital audio information into an analog audio signal for output, and is also configured to convert analog audio input into a digital audio signal. The audio module **170** may be further configured to encode and decode an audio signal.

[0229] The speaker **170**A, also referred to as a “loudspeaker”, is configured to convert an audio electrical signal into a sound signal.

[0230] The receiver **170**B, also referred to as an “earpiece”, is configured to convert an audio

electrical signal into a sound signal.
[0231] The microphone **170**C, also referred to as a “mike” or a “mic”, is configured to convert a sound signal into an electrical signal. For example, in some embodiments, the microphone **170**C may collect a voice of the user and send the voice to the processor **110**. When detecting that the voice of the user includes voice data such as “image capturing”, “screenshot taking”, and “screen recording”, the processor **110** may trigger the content-obtaining method in embodiments of this application.
[0232] The headset jack **170**D is configured to connect to a wired headset.
[0233] The pressure sensor **180**A is configured to sense a pressure signal and convert the pressure signal into an electrical signal. In some embodiments, the pressure sensor **180**A may be disposed on the display **194**. There are a plurality of types of pressure sensors **180**A, such as a resistive pressure sensor, an inductive pressure sensor, and a capacitive pressure sensor. The capacitive pressure sensor may include at least two parallel plates made of conductive materials. When a force is applied to the pressure sensor **180**A, capacitance between electrodes changes. The mobile phone **10** determines pressure strength based on a change of the capacitance. When a touch operation is performed on the display **194**, the mobile phone **10** detects intensity of the touch operation by using the pressure sensor **180**A. The mobile phone **10** may also calculate a touch location based on a detection signal of the pressure sensor **180**A.
[0234] The acceleration sensor **180**E may detect magnitude of accelerations of the mobile phone **10** in various directions (usually on three axes). When the mobile phone **10** is still, a value and a direction of gravity may be detected. The acceleration sensor **180**E may be further configured to: recognize a posture of the electronic device, determine whether the mobile phone **10** is in a landscape mode, and determine a status of an application such as a pedometer.
[0235] The ambient light sensor **180**L is configured to sense ambient light brightness. The mobile phone **10** may adaptively adjust brightness of the display **194** based on the sensed ambient light brightness. The ambient light sensor **180**L may also be configured to automatically adjust white balance during photographing. The ambient light sensor **180**L may also cooperate with the optical proximity sensor **180**G, to detect whether the mobile phone **10** is in a pocket, to avoid an accidental touch.
[0236] The fingerprint sensor **180**H is configured to capture a fingerprint. The mobile phone **10** may use a feature of the captured fingerprint to implement fingerprint-based unlocking, application lock access, fingerprint-based photographing, fingerprint-based call answering, and the like.
[0237] The touch sensor **180**K is also referred to as a “touch device”. The touch sensor **180**K may be disposed on the display **194** and the touch sensor **180**K and the display **194** constitute a touchscreen, which is also referred to as a “touch screen”. The touch sensor **180**K is configured to detect a touch operation performed on or near the touch sensor. The touch sensor may transfer the detected touch operation to the application processor to determine a type of a touch event. A visual output related to the touch operation may be provided on the display **194**. In some other embodiments, the touch sensor **180**K may alternatively be disposed on a surface of the mobile phone **10** at a location different from that of the display **194**. For example, in some embodiments, when detecting an operation performed by the user on the touchscreen, the touchscreen may send a touch event to the processor **110** so that the processor **110** can recognize a gesture of the user to trigger the content-obtaining method provided in embodiments of this application.
[0238] The bone conduction sensor **180**M may obtain a vibration signal. In some embodiments, the bone conduction sensor **180**M may obtain a vibration signal of a vibration bone of a human vocal-cord part. The bone conduction sensor **180**M may also be in contact with a body pulse to receive a blood pressure beating signal.
[0239] The button **190** includes a power button, a volume button, and the like. The button **190** may be a mechanical button or may be a touch button. The mobile phone **10** may receive a button input and generate a key signal input related to a user setting and function control of the mobile phone

10. In some embodiments, when detecting a double-press operation performed by the user on the volume button, the processor **110** may trigger the content-obtaining method provided in embodiments of this application.

[0240] The motor **191** may generate a vibration prompt.

[0241] The indicator **192** may be an indicator light, and may be configured to indicate a charging status and a power change, or may be configured to indicate a message, a missed call, a notification, and the like.

[0242] The SIM card interface **195** is configured to connect to a SIM card.

[0243] It may be understood that the structure of the mobile phone **10** shown in embodiments of this application does not constitute a limitation on the mobile phone **10**. In some other embodiments of this application, the mobile phone **10** may include more or fewer components than those shown in the figure, or a combination of some components, or splits from some components, or a different component layout. The components shown in the figure may be implemented by hardware, software, or a combination of software and hardware.

[0244] Embodiments of mechanisms disclosed in this application may be implemented by hardware, software, firmware, or a combination of these implementation methods. Embodiments of this application may be implemented as a computer program or program code executed in a programmable system. The programmable system includes at least one processor, a storage system (including a volatile memory, a non-volatile memory, and/or a storage element), at least one input device, and at least one output device.

[0245] The program code may be applied to input instructions to execute the functions described in this application and generate output information. The output information may be applied to one or more output devices in a known manner. For a purpose of this application, a processing system includes any system having a processor such as a digital signal processor (DSP), a microcontroller, an application-specific integrated circuit, or a microprocessor.

[0246] The program code may be implemented in a high-level programming language or an object-oriented programming language to communicate with the processing system. The program code may alternatively be implemented in an assembly language or a machine language when required. The mechanisms described in this application are not limited to the scope of any particular programming language. In any case, the language may be a compiled language or an interpretive language.

[0247] In some cases, the disclosed embodiments may be implemented in hardware, firmware, software, or any combination thereof. The disclosed embodiments may alternatively be implemented as instructions that are carried or stored on one or more transitory or non-transitory machine-readable (for example, computer-readable) storage media and that may be read and executed by one or more processors. For example, the instructions may be distributed through a network or another computer-readable medium. Therefore, the machine-readable medium may include any mechanism for storing or transmitting information in a machine (for example, a computer) readable form, including but not limited to a floppy disk, a compact disc, an optical disc, a read-only memory (CD-ROMs), a magnetic optical disk, a read-only memory (ROM), a random access memory (RAM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), a magnetic card, an optical card, a flash memory, or a tangible machine-readable memory used to transmit information (for example, a carrier, an infrared signal, or a digital signal) by using a propagating signal in an electrical, optical, acoustic, or another form over the internet. Therefore, the machine-readable medium includes any type of machine-readable medium suitable for storing or transmitting an electronic instruction or information in a machine (namely, computer)-readable form.

[0248] In the accompanying drawings, some structural or method features may be shown in a particular arrangement and/or order. However, it should be understood that such a particular arrangement and/or order may not be needed. In some embodiments, these features may be

arranged in a manner and/or an order different from that shown in the descriptive accompanying drawings. In addition, inclusion of the structural or method features in a particular figure does not imply that such features are needed in all embodiments, and in some embodiments, these features may not be included or may be combined with other features.
[0249] It should be noted that all units/modules mentioned in device embodiments of this application are logical units/modules. Physically, one logical unit/module may be one physical unit/module, may be a part of one physical unit/module, or may be implemented by a combination of a plurality of physical units/modules. Physical implementations of these logical units/modules are not the most important, and a combination of functions implemented by these logical units/modules is a key to resolve the technical problem provided in this application. In addition, to highlight an innovative part of this application, a unit/module that is not closely related to resolving the technical problem provided in this application is not introduced in the foregoing device embodiments of this application. This does not mean that there are no other units/modules in the foregoing device embodiments.
[0250] It should be noted that in examples and this specification of this patent, relational terms such as first and second are used only to differentiate an entity or operation from another entity or operation, and do not require or imply that any actual relationship or sequence exists between these entities or operations. Moreover, the term “include” or any other variant thereof is intended to cover a non-exclusive inclusion, so that a process, a method, an article, or a device that includes a list of elements not only includes those elements but also includes other elements that are not expressly listed, or further includes elements inherent to such a process, method, article, or device. Without further limitations, an element limited by “include a/an” does not exclude other same elements existing in the process, the method, the article, or the device that includes the element.
[0251] Although this application has been illustrated and described with reference to some preferred embodiments of this application, a person of ordinary skill in the art should understand that various changes may be made to this application in form and detail without departing from the spirit and scope of this application.

Claims

1. A content-obtaining method for an electronic device, the method comprising: detecting a content-obtaining instruction of a user for a display interface on a screen of the electronic device; and generating display content data corresponding to the display interface by redrawing the first security control in a first redrawing manner to generate a display content data comprising first redrawn data, in response to detecting the content-obtaining instruction and the display interface comprising a first security control.
2. The method according to claim 1, wherein when a security control identifier corresponding to the first security control is valid, the electronic device determines that the display interface comprises the first security control.
3. The method according to claim 2, wherein the security control identifier of the first security control is set to valid when a layout file of a first application corresponding to the first security control is loaded.
4. The method according to claim 1, wherein the first redrawing manner is obtained from a layout file of a first application corresponding to the first security control.
5. The method according to claim 1, wherein the first redrawing manner comprises at least one of the following: content clearing, filling, content replacing, hiding, or re-layout.
6. The method according to claim 5, wherein when the first redrawing manner is the content replacing, the redrawing the first security control in the first redrawing manner comprises: obtaining a data type of user sensitive data in the first security control; and replacing the user sensitive data with preset data whose data type is the same as that of the user sensitive data.

7. The method according to claim 5, wherein when the first redrawing manner is the re-layout, the redrawing the first security control in the first redrawing manner comprises: deleting a placeholder of the first security control in a current layout of the first application corresponding to the first security control; and re-laying out another control in the current layout.
8. The method according to claim 5, wherein when the first redrawing manner is the content clearing, the redrawing the first security control in the first redrawing manner comprises: clearing user sensitive data in the first security control and retaining a placeholder of the first security control in a current layout of the first application corresponding to the first security control.
9. The method according to claim 5, wherein when the first redrawing manner is the filling, the redrawing the first security control in the first redrawing manner comprises: filling an area in which user sensitive data in the first security control is located with any one or more of a preset color, a pattern, or a mosaic.
10. The method according to claim 5, wherein when the first redrawing manner is the hiding, the redrawing the first security control in the first redrawing manner comprises: hiding the first security control and retaining a placeholder of the first security control in a current layout of the first application corresponding to the first security control.
11. The method according to claim 1, wherein the display interface further comprises a second security control, and the display content data comprises second redrawn data obtained by redrawing the second security control in a second redrawing manner.
12. The method according to claim 1, wherein the content-obtaining instruction comprises a screenshot-taking instruction or a screen-recording instruction, and the display content data comprises a screenshot image or a video.
13. The method according to claim 1, wherein an operating system of the electronic device comprises a graphics drawing module; and when the display interface comprises the first security control, the graphics drawing module generates the display content data corresponding to the content-obtaining instruction.
14. A non-transitory computer-readable media storing computer instructions that configure at least one processor, upon execution of the instructions, to perform the following steps: detecting a content-obtaining instruction of a user for a display interface on a screen of the electronic device; and generating display content data corresponding to the display interface by redrawing the first security control in a first redrawing manner to generate a display content data comprising first redrawn data, in response to detecting the content-obtaining instruction and the display interface comprising a first security control.
15. The computer-readable storage medium according to claim 14, wherein when a security control identifier corresponding to the first security control is valid, the electronic device determines that the display interface comprises the first security control.
16. The computer-readable storage medium according to claim 15, wherein the security control identifier of the first security control is set to valid when a layout file of a first application corresponding to the first security control is loaded.
17. The computer-readable storage medium according to claim 14, wherein the first redrawing manner is obtained from a layout file of a first application corresponding to the first security control.
18. The computer-readable storage medium according to claim 14, wherein the first redrawing manner comprises at least one of the following: content clearing, filling, content replacing, hiding, or re-layout.
19. The computer-readable storage medium according to claim 18, wherein when the first redrawing manner is the content replacing, the redrawing the first security control in the first redrawing manner comprises: obtaining a data type of user sensitive data in the first security control; and replacing the user sensitive data with preset data whose data type is the same as that of the user sensitive data.

20. The computer-readable storage medium according to claim 18, wherein when the first redrawing manner is the re-layout, the redrawing the first security control in the first redrawing manner comprises: deleting a placeholder of the first security control in a current layout of the first application corresponding to the first security control; and re-laying out another control in the current layout.